Thermally-Managed Solid-State Battery Module with Embedded Micro-Sensors

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Team Name: Optimus Prime

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Problem Statement

Thermally-Managed Solid-State Battery Module with Embedded Micro-Sensors.

- During DC fast-charging, cells heat unevenly and hot spots form at cell interfaces and tabs, where pack-level cooling cannot reach in time.
- Many packs rely on sparse, module-level temperature sensing, so local cell-to-cell differences can go unnoticed until they grow large (Liu et al., 2019).
- The aim is a battery module that links per-cell micro-sensing to targeted cooling. The first demonstrator uses conventional lithium-ion cells to validate the thermal-control architecture; solid-state compatibility is a future research objective, not a demonstrated capability.

Problem Understanding

- **Who is affected:** EV owners and fleet operators (cabs, delivery vehicles, buses) that fast-charge frequently, and the manufacturers who design and validate battery thermal systems.
- **Root causes:** uneven current distribution, differences in cell internal resistance, and limited heat paths between cells; resistive heating scales with current squared (I^2R), so faster charging sharply raises heat.
- **Consequences:** cells age unevenly and the weakest cell limits pack life; the BMS throttles charging power as a precaution, and the risk of thermal runaway rises (Zhang et al., 2022).
- **Why now:** high-power DC charging (of the order of 150 kW) is becoming common, and a commonly cited operating window for conventional lithium-ion is roughly 25–40 °C, with cell-to-cell spread kept within a few degrees as an engineering target; the exact limits depend on cell chemistry, construction and charge rate (Akbarzadeh et al., 2021).

Existing Gaps & Challenges

- **Sparse sensing:** where sensing is placed only at selected module locations, hot spots at cell interfaces and tabs can be detected late; cell-level monitoring remains an active research area (Liu et al., 2019).
- **Uniform cooling:** cold plates and cooling ribbons tend to cool the whole pack with little local control, and air cooling is limited at fast-charge heat loads (Zhang et al., 2022; Xia et al., 2017).
- **Passive buffering limits:** phase change materials help with uniformity but suffer from low thermal conductivity, latent-heat saturation and leakage (Goli et al., 2014).
- **Largely reactive control:** many conventional systems raise cooling toward peak power once thresholds are approached; the proposed design should be compared against a clearly defined baseline controller.
- **Opportunity:** low-cost systems that tightly couple per-cell sensing to targeted cooling in real time are not yet common in production (subject to market and prior-art review).

Our Proposed Solution

- **Core idea:** a battery-module enclosure that combines a mini-channel liquid cold plate, intended to be divided into independently addressable cooling zones, with a per-cell temperature micro-sensor array.
- **How it works:** the micro-sensors map cell temperatures in near real time; a controller is intended to route more coolant to the hottest zones while keeping the rest at a low-power baseline.
- **Hybrid buffering:** a graphene-enhanced phase change material layer is proposed between the cells and the cold plate to absorb short, high-rate thermal peaks (Goli et al., 2014; Hussain et al., 2018).

Central testable proposition: can closed-loop, zone-specific coolant control, informed by distributed sensing and supported by a PCM buffer, reduce peak temperature and total cooling energy versus conventional uniform cooling under the same heat load?

Technical Approach



To be specified: sensor sampling rate, control interval, thresholds and hysteresis, flow feedback, and fail-safe behaviour on sensor or valve failure. Independent zonal control also requires a defined hydraulic schematic, valve specifications and pump curve, since opening one valve can change flow in other branches.

Tech Stack

Hardware: 4–6 Li-ion cells (stand-in for solid-state), mini-channel cold plate, thermistor/RTD micro-sensor array, micro-valves, film heaters to emulate fast-charge heat

Software: SolidWorks/Fusion CAD; ANSYS/COMSOL thermal–CFD

Controller: ESP32/Arduino with zone-wise logic; small pump, tubing and water–glycol coolant

Modelling: Python/MATLAB data logging and a lumped thermal model

Innovation & Uniqueness

Figures below are published results under each study's specific conditions; whether they transfer to the proposed module is still to be tested.

- **Per-cell, in-situ sensing** instead of sparse module-level readings, to detect hot spots earlier (Liu et al., 2019).
- **Zonal, micro-valved cooling** aims to direct coolant where heat is; a hybrid liquid-plus-PCM cold plate cut pumping energy by up to about 30% versus a conventional aluminium plate in its own design and test conditions (Akbarzadeh et al., 2021).
- **Graphene-enhanced PCM buffer:** graphene raised PCM thermal conductivity by more than two orders of magnitude as a material property (Goli et al., 2014); graphene-coated nickel foam with paraffin lowered cell surface-temperature rise by about 17% relative to nickel foam at a 1.7 A discharge (Hussain et al., 2018).
- **Intended differentiator (to be validated):** one low-cost module that couples per-cell sensing to targeted cooling; novelty is to be established against defined baselines and a prior-art review.

Feasibility & Scalability

- **Feasibility:** the demonstrator can be built from off-the-shelf parts with lab access and a mentor; prior work reports distributed forced convection as a cost-effective route to uniform temperature (Karimi and Li, 2013) and hybrid liquid-plus-PCM plates tested experimentally (Akbarzadeh et al., 2021).
- **Validation path:** first in thermal–CFD simulation, then on a small 4–6 cell module, compared against conventional uniform-flow cooling at the same heat input, measuring cooling energy and not only temperature.
- **Success criteria (provisional targets):** tighter cell-to-cell spread and lower peak temperature at equal or lower cooling energy than the baseline.
- **Scalability:** the zoned module is intended to be modular; a small demonstrator does not by itself establish vehicle-pack scalability, durability, serviceability or safety, which require further work.
- **Risks and mitigations:** per-cell wiring cost and failure points (multiplexed or flexible printed-circuit arrays); PCM leakage (shape-stabilised or composite PCM); coolant sealing and service access.

Cost for Product (Revised, Indicative)

Component / expense	Estimate
4–6 cells, holders and enclosure	Rs. 4,000–6,000
Mini-channel cold plate: fabrication and leak testing	Rs. 7,000–12,000
Pump, tubing, fittings and coolant	Rs. 3,000–5,000
Temperature sensor array, ESP32 and electronics	Rs. 4,000–7,000
Micro-valves and valve-driver circuits	Rs. 2,000–4,000
Graphene-enhanced PCM and containment	Rs. 3,000–6,000
Film heaters, wiring and consumables	Rs. 2,000–3,000
Power supply, fuses and safety provisions	Rs. 2,000–4,000
Reference sensors, flow measurement and data acquisition	Rs. 4,000–8,000
Assembly, calibration and testing	Rs. 3,000–6,000
Subtotal	**Rs. 34,000–61,000**
Contingency (approximately 10%)	Rs. 3,400–6,100
Estimated total	**Rs. 37,400–67,100**

Proposed working budget

Rs. 52,000

for an instrumented 4–6-cell laboratory demonstrator

Hardware and thermal-management components: Rs. 30,000
Instrumentation, calibration and testing: Rs. 14,000
Assembly, consumables and contingency: Rs. 8,000

Planning range approximately Rs. 40,000–67,000.

A preliminary engineering estimate, not a verified supplier quotation. Rs. 52,000 is a selected planning budget, not the midpoint of the range; the final cost depends on supplier quotations, access to shared laboratory equipment and the selected PCM formulation.

Use Cases (compulsorily 3 use cases must be there)



- **Use case 1 — Fast-charging fleet vehicles:** taxis, delivery vans and buses that fast-charge several times a day; targeted cooling is intended to hold the cell-to-cell spread tight across repeated DC fast charges, with the aim of extending pack life and reducing downtime (potential outcomes, to be validated).
- **Use case 2 — Manufacturer validation rig:** battery-pack makers could use the per-cell sensing and zonal cooling module as a test platform to study thermal behaviour of new cell formats, including pilot solid-state cells.
- **Use case 3 — High-ambient, high-C-rate operation:** vehicles in hot climates or at high discharge rates; proactive zonal cooling aims to limit local hot spots and reduce precautionary throttling of charge or discharge power.

Impacts and Benefits

The items below are expected benefits contingent on validation, not outcomes demonstrated by the prototype.

- **For users:** aims to enable faster charging and longer battery life, which would lower the cost per kilometre for drivers and fleets.
- **For manufacturers:** potential for fewer warranty claims from uneven aging, and a thermal-control architecture that could inform future solid-state packs.
- **Safety framing:** the system aims to improve thermal monitoring and temperature control; lowering peak temperature and improving uniformity supports safety but does not, by itself, demonstrate prevention of thermal runaway.
- **Projected targets (to be validated):** cell-to-cell spread within about 5 °C and peak temperature below about 40 °C during fast-charge-like loads, at equal or lower cooling energy than uniform cooling (cf. Akbarzadeh et al., 2021).

Sustainable Development Goals (SDGs)

SDG Number	Target	Goal Name	Connection
SDG 7	Target 7.3	Affordable and Clean Energy	Aims to cut parasitic pack cooling energy
SDG 9	Target 9.4	Industry, Innovation and Infrastructure	Efficient battery thermal technology
SDG 11	Target 11.6	Sustainable Cities and Communities	Indirect: depends on EV adoption and grid mix
SDG 13	Target 13.2	Climate Action	Indirect: supports transport decarbonisation

SDG links indicate strategic alignment, not a measured environmental benefit.

Individual Member Contribution

Member	Contribution or responsibility
G A Yuvan	Cost research, use-case identification and impact study
Manu K	Problem research and understanding, SDG identification
Kousik Kumar	Research-gap identification, solution proposal and refining the technical approach
Harshitha Kothandam Sasikumar	Innovation possibilities and feasibility study

Yukti ID

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